

Measures of Central Tendency (Detailed)

Measures of Central Tendency are statistical measures that identify the **central or typical value** of a dataset. They summarize a large set of data with a single representative value, making it easier to understand and compare datasets.

In **Data Science**, measures of central tendency are often the **first step in Exploratory Data Analysis (EDA)** because they provide a quick understanding of the data's center.

Definition

Measures of Central Tendency are statistical methods used to determine the central, average, or most representative value of a dataset.

They answer the question:

"What is the typical value of this dataset?"

Why are Measures of Central Tendency Important?

They help us to:

- Summarize large datasets.
 - Compare different datasets.
 - Understand the distribution of data.
 - Detect unusual values (outliers).
 - Prepare data for Machine Learning models.
 - Fill missing values in datasets.
 - Make business decisions based on average behavior.
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Types of Measures of Central Tendency

There are **three main measures**:

1. **Mean (Arithmetic Mean)**
 2. **Median**
 3. **Mode**
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1. Mean (Arithmetic Mean)

The **Mean** is the average of all values in a dataset.

It is calculated by adding all the values and dividing by the total number of observations.

FORMULA

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

- **$\sum X$** = Sum of all observations
- **N** = Total number of observations

```
genui{"descriptive_statistics_sampling_learning_block_staging":  
{"type_id":"ARITHMETIC_MEAN"}}□
```

EXAMPLE 1

Marks of five students:

60, 70, 80, 90, 100

STEP 1

Find the sum

$$60 + 70 + 80 + 90 + 100 = 400$$

STEP 2

Divide by the total number of observations

$$400 / 5 = 80$$

ANSWER

Mean = 80

EXAMPLE 2

Daily temperatures (°C)

28, 30, 32, 31, 29

Mean

$$(28 + 30 + 32 + 31 + 29) / 5 = 30^{\circ}\text{C}$$

ADVANTAGES OF MEAN

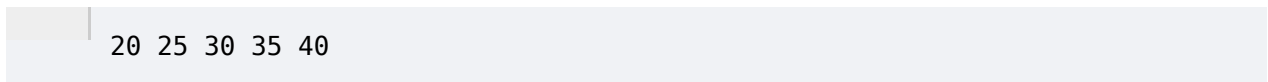
- Easy to calculate.
 - Uses every observation.
 - Most commonly used average.
 - Useful in mathematical calculations.
 - Preferred in Machine Learning algorithms.
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DISADVANTAGES OF MEAN

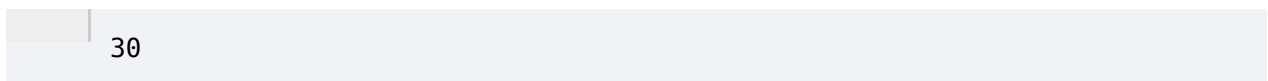
- Highly affected by **outliers**.
 - Not suitable for skewed data.
 - Cannot be used with categorical data.
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EXAMPLE OF OUTLIER EFFECT

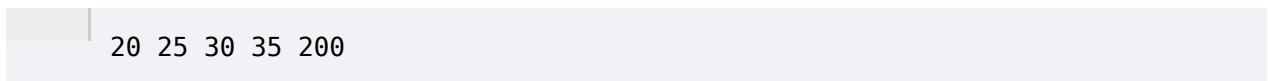
Dataset A



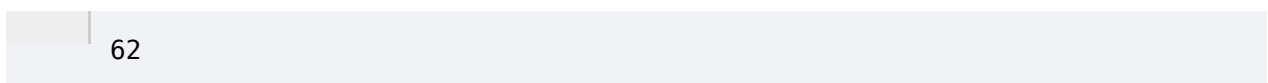
Mean



Dataset B



Mean



Although most values are near 30, the mean becomes **62** because of the outlier (200).

2. Median

The **Median** is the **middle value** of a dataset after arranging the data in ascending or descending order.

Unlike the mean, it is **not affected much by outliers**, making it useful for skewed data.

STEPS TO FIND THE MEDIAN

1. Arrange the data in order.
 2. Count the observations.
 3. Find the middle value.
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CASE 1: ODD NUMBER OF OBSERVATIONS

Example

5, 8, 12, 15, 18

Middle value = **12**

Median = 12

CASE 2: EVEN NUMBER OF OBSERVATIONS

Example

10, 20, 30, 40

Middle two values:

20 and 30

Median

$(20 + 30) / 2 = 25$

Median = 25

ADVANTAGES OF MEDIAN

- Not affected by outliers.
 - Best for skewed distributions.
 - Suitable for ordinal data.
 - Easy to understand.
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DISADVANTAGES OF MEDIAN

- Does not use all observations.
 - Less suitable for advanced mathematical analysis.
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EXAMPLE

Monthly incomes

₹20,000

₹25,000

₹30,000

₹35,000
₹5,00,000

Mean

₹1,22,000

Median

₹30,000

Here, the **median better represents the typical income** because one extremely high income (₹5,00,000) pulls the mean upward.

Mean vs. Median (Effect of Outlier)

```
□genui□{"descriptive_statistics_sampling_learning_block_staging":  
{"type_id":"MEAN_VS_MEDIAN"}}□
```

Notice that the **mean shifts significantly when an extreme value is introduced, while the median stays much more stable.**

3. Mode

The **Mode** is the value that appears **most frequently** in a dataset.

It is the only measure of central tendency that can be used with **categorical (non-numeric) data**.

EXAMPLE 1

Dataset

10, 20, 20, 30, 40

20 appears twice.

Mode = 20

EXAMPLE 2

2, 2, 2, 4, 4, 5, 6

Mode = 2

EXAMPLE 3 (NO MODE)

10, 20, 30, 40

Every value appears only once.

No Mode

EXAMPLE 4 (TWO MODES)

5, 5, 8, 8, 10

Modes:

5 and 8

This is called **Bimodal** data.

EXAMPLE 5 (MULTIPLE MODES)

1, 1, 2, 2, 3, 3

Modes:

1, 2, 3

This is called **Multimodal** data.

ADVANTAGES OF MODE

- Easy to identify.
 - Works with categorical data.
 - Not affected by extreme values.
 - Useful in market research (e.g., most purchased product).
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DISADVANTAGES OF MODE

- May not exist.
 - May have more than one mode.
 - Does not use all observations.
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Comparison of Mean, Median, and Mode

Feature	Mean	Median	Mode
Definition	Average of all values	Middle value	Most frequent value
Uses all observations	✓ Yes	✗ No	✗ No
Affected by outliers	✓ Yes	✗ No (or very little)	✗ No
Suitable for categorical data	✗ No	✗ No	✓ Yes
Mathematical calculations	✓ Excellent	Limited	Limited
Best for skewed data	✗ No	✓ Yes	Sometimes

Choosing the Right Measure

Situation	Best Measure
Symmetric data	Mean

without outliers	
Skewed data (salary, income, house prices)	Median
Most common category or value	Mode
Product popularity	Mode
Student average marks	Mean
Household income	Median

Applications in Data Science

Measures of central tendency are used in:

- **Exploratory Data Analysis (EDA):** Understanding the center of the data.
 - **Missing value imputation:** Replacing missing values with the mean or median.
 - **Business analytics:** Calculating average sales, revenue, or customer spending.
 - **Healthcare:** Finding average patient age or common symptoms.
 - **Finance:** Analyzing salaries, stock prices, and loan amounts.
 - **Machine Learning:** Summarizing features before model training.
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Example in Python

```
import statistics

data = [10, 20, 20, 30, 40]

print("Mean :", statistics.mean(data))
print("Median :", statistics.median(data))
print("Mode :", statistics.mode(data))
```

Output

```
Mean : 24
Median : 20
Mode : 20
```

Summary

Measure	Meaning	Best Used When
Mean	Average of all values	Data is symmetric and has no significant outliers
Median	Middle value	Data is skewed or contains outliers
Mode	Most frequently	Finding the most common

	occurring value	value or category
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KEY TAKEAWAY

- **Mean** is the most widely used average but is sensitive to extreme values.
- **Median** is the best choice for skewed data because it is resistant to outliers.
- **Mode** identifies the most common value and is the only central tendency measure applicable to categorical data.
- In **Data Science**, all three measures are used together during **Exploratory Data Analysis (EDA)** to gain a comprehensive understanding of the dataset before building predictive models.